

Name:R.RAGHUVeer

Roll Number:25220030225

1:

GNU nano 8.7.1

```
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main() {
    int server_fd, client_fd;
    struct sockaddr_un address;
    char buffer[100];

    // Create socket
    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    // Set socket address
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    // Remove old socket
    unlink(SOCKET_PATH);

    // Bind socket to path
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    // Wait for client
    listen(server_fd, 5);

    printf("Server waiting...\n");

    // Accept client
    client_fd = accept(server_fd, NULL, NULL);

    // Receive message
    memset(buffer, 0, sizeof(buffer));
    read(client_fd, buffer, sizeof(buffer) - 1);

    printf("Message from client: %s\n", buffer);

    // Close sockets
    close(client_fd);
    close(server_fd);

    // Remove socket file
    unlink(SOCKET_PATH);

    return 0;
}
```

OUTPUT:

```
raghu@Raghu:~$ nano socketserver.c
raghu@Raghu:~$ gcc socketserver.c -o
raghu@Raghu:~$ ./socketserver
Server waiting...
Message from client: Hello Server
raghu@Raghu:~$ |
```

2:

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/socket.h>  
#include <sys/un.h>  
#include <string.h>  
  
#define SOCKET_PATH "/tmp/my_socket"  
  
int main()  
{  
    int client_fd;  
    struct sockaddr_un address;  
  
    // Create socket  
    client_fd = socket(AF_UNIX, SOCK_STREAM, 0);  
  
    // Set socket address  
    address.sun_family = AF_UNIX;  
    strcpy(address.sun_path, SOCKET_PATH);  
  
    // Connect to server  
    connect(client_fd,  
            (struct sockaddr *)&address,  
            sizeof(address));  
  
    // Send message  
    write(client_fd, "Hello Server", 12);  
  
    printf("Message sent to server.\n");  
  
    close(client_fd);  
  
    return 0;  
}  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T E  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J J
```

OUTPUT:

```
raghu@Raghu:~$ nano socketclient.c
raghu@Raghu:~$ gcc socketclient.c -
raghu@Raghu:~$ ./socketclient
Message sent to server.
raghu@Raghu:~$ nano socketclient.c
raghu@Raghu:~$ gcc socketclient.c -
raghu@Raghu:~$ ./socketclient
Message sent to server.
```